Ashkan Nikfajram

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EDUCATION

Bachelor of Science, Data Science (GPA: 3.6) May 2025
San Jose State University, San Jose, CA

RELEVANT SKILLS

Languages: C++, Java, Python, JavaScript(React), TypeScript, HTML, CSS, shadcn/ui, MUI **| Agents & Protocols:** Model Context Protocol (MCP), LangChain, LangGraph, LangSmith **| ML Frameworks:** Scikit-Learn, PyTorch, TensorFlow, Keras, Vaex **| Data Libraries:** Pandas, Seaborn, Matplotlib, Plotly, BeautifulSoup, Dash, Marimo, Jupyter **| Databases & ORMs:** SQLite, MySQL, SQLAlchemy, Firebase RealTime DB, FireStore DB **| Cloud & DevOps:** DockerHub, GitHub, Kubernetes, GCP (Compute Engine, Cloud Composer, Pub/Sub, BigQuery), Vercel, Render, Azure (Cloud Storage, VMs) **| APIs & Backend:** REST APIs, FastAPI,Flask, Pydantic, Auth0 **| Workflow & Orchestration:** Dagster, Prefect, Terraform

WORK EXPERIENCE

Software Engineer, Temp, Stanford School of Medicine Genomic Lab, Stanford, CA Sep 2025 - Present

- Engineered full-stack features for an AI/LLM-based genomic research web application enabling regulatory-genome studies across diverse cell types.
- Designed and optimized ML preprocessing workflows, reducing pipeline failure rates from ~20% to 1–2% by improving validation logic, error handling, logging, and modular pipeline design.
- Implemented real-time genomic variant scoring using the AlphaGenome deep learning model, enabling on-the-fly annotation and scoring of thousands of variants per request for downstream regulatory impact analysis.
- Designed and maintained scalable data–model interaction workflows across multi-cloud environments using Terraform for infrastructure as code and Prefect and other orchestration tools, ensuring reproducible and fault-tolerant pipeline execution.
- Built automated validation and preprocessing pipelines for genomic variant data (TSV/FASTA), including region filtering, schema validation, and coding/non-coding classification to ensure data integrity prior to model inference.
- Deployed a JupyterHub server to manage user notebooks (Marimo), allowing researchers to import and analyze the model's result data easily.

IT Consultant, Robert Half Technology, San Francisco, CA May 2019 - Apr 2020

- Managed clients' servers and provided user support to enhance office efficiency and productivity
- Developed and implemented Python and Bash scripts tailored to client IT projects, automating routine tasks such as backup and mapping network drive scripts, and updating and pushing Cisco network configurations
- Engineered comprehensive dashboards for IT departments using JavaScript/HTML/CSS to centralize access to IT resources and actions, such as to manage inventory or monitor server health

PROJECT EXPERIENCE

GeneScope: MultiModal Deep Learning for Breast Cancer Biomarker Discovery & Prognosis Jan 2025 - May 2025
https://gene-scope-liard.vercel.app/ , *https://github.com/ANikfarjam/GeneScope*

- Built a full-stack web application allowing users to classify breast cancer stages using a Multi Modal DNN with intermediate fusion, MLP, and CatBoost models optimized for imbalanced datasets
- Combined statistical and ML methods (Analytical Hierarchical Process, Cox Hazard Regression, Gradient Boosting Trees) to analyze biomarkers and prognosis, with findings visualized in an interactive dashboard
- Integrated a LangChain-based chatbot powered by Pinecone vector search to answer user questions about gene roles, clinical terms, and biological concepts based on indexed documents and lecture notes

StairCase: Multiplayer Game with AI Minigames Jan 2025 - May 2025
https://stair-case.vercel.app/ , *https://github.com/ANikfarjam/StairCase*

- Built a cross-platform Snakes & Ladders-style game using Pygame, Flask, Firebase Auth, and Firestore DB with real-time multiplayer and party system support
- Designed and deployed a full-stack architecture (React frontend on Vercel, Flask backend on NorthFlank via Docker) with modular API routes using Flask Blueprints
- Integrated LangChain agents, utilizing Mistral AI LLM, to power dynamic trivia and hangman minigames, enhancing gameplay with LLM-generated content

X-Ray Image Classification Oct 2024 - Dec 2024
https://github.com/ANikfarjam/X-ray-Image_classification_CNNRNN

- Trained a Convolutional Neural Network along with a bidirectional LSTM Recurrent Neural Network to classify X-ray images based on location and identify fractured versus non-fractured bones
- Utilized Keras Tuner NAS(Network Architecture Search), to optimize hyperparameters and network structure, achieving a test accuracy of 89.5%